NAN MUDHALVAN

PROJECT REPORT

COURIER MANAGEMENT SYSTEM

A Web-Based Courier Booking, Tracking and Delivery Management Application

Submitted by

[STUDENT NAME]

B.Sc. Computer Science

Bharathi Women’s Arts and Science College

Academic Year 2026–2027

CERTIFICATE

This is to certify that the project entitled “Courier Management System” is a bonafide project work carried out as part of the Nan Mudhalvan academic/project activity by [STUDENT NAME], B.Sc. Computer Science, Bharathi Women’s Arts and Science College, during the academic year 2026–2027.

Project Guide	Head of the Department	Principal
Signature: _____	Signature: _____	Signature: _____

DECLARATION

I hereby declare that the project entitled “Courier Management System” is my academic project work prepared as part of the Nan Mudhalvan programme. The project has been developed for educational purposes to demonstrate the application of software development, database management and web technologies in solving a practical courier-management problem.

I further declare that the information and design presented in this report are prepared for academic use and that the final implementation may be modified according to the technologies and requirements selected during development.

ACKNOWLEDGEMENT

I express my sincere gratitude to the management, Principal, Head of the Department and faculty members of Bharathi Women's Arts and Science College for providing the support and guidance required to complete this project.

I also thank the Nan Mudhalvan programme and the learning resources that helped me understand modern software development concepts and apply them to a practical project. Finally, I thank my classmates, friends and family members for their encouragement and support throughout the project work.

ABSTRACT

The Courier Management System is a web-based application developed to computerize and simplify the major activities involved in courier service management. Courier organizations handle a large amount of information related to customers, parcels, bookings, tracking numbers, shipment status, delivery personnel, charges and delivery confirmation. Managing these details manually can consume time and may lead to errors, duplicate records and difficulty in tracking shipments.

The proposed system provides a centralized platform where courier bookings can be created, shipment information can be stored, tracking IDs can be generated and delivery status can be updated. The administrator can manage users and courier records, while customers can check their shipment information and track the current status of their parcels.

The project demonstrates the practical use of web development and database management concepts. The system can be extended in the future with online payment, GPS tracking, automated notifications, mobile applications and analytics.

EXECUTIVE SUMMARY

The Courier Management System is designed around a simple idea: every courier should have a unique identity and a clear status from booking to delivery. The application maintains this information in a structured database so that authorized users can retrieve and update it efficiently.

Area	Proposed Solution
Customer Management	Digital storage and management of customer details
Courier Booking	Create and store shipment bookings
Tracking	Unique tracking ID for every courier
Status Management	Update shipment progress at different stages
Delivery	Record delivery confirmation
Reports	View and summarize courier information
Security	Role-based access to important functions

TABLE OF CONTENTS

1. Introduction

2. Background of the Project

3. Problem Statement

4. Existing System

5. Proposed System

6. Objectives

7. Scope

8. Feasibility Study

9. Requirements Analysis

10. System Users and Roles

11. System Modules

12. System Architecture

13. Data Flow Diagram

14. Entity Relationship Design

15. Database Design

16. Functional Requirements

17. Non-Functional Requirements

18. User Interface Design

19. System Workflow

20. Algorithms and Logic

21. Testing Strategy

22. Test Cases

23. Security Considerations

24. Advantages

25. Limitations

26. Future Enhancements

27. Expected Outcomes

28. Conclusion

29. References

1. INTRODUCTION

Courier services are an important part of modern communication and e-commerce. Individuals and organizations regularly send documents, parcels, products and other items from one location to another. As the number of courier transactions increases, maintaining accurate information becomes important for both the service provider and the customer.

A manual system may depend on registers, spreadsheets, phone calls and paper receipts. Although these methods can work for a small number of transactions, they become difficult to manage when the number of customers and shipments increases.

The Courier Management System addresses this problem by providing a centralized digital system for recording courier details, generating tracking information, updating shipment status and maintaining delivery records.

1.1 Need for the Project

To reduce dependency on paper-based records.

To make courier information easier to search and retrieve.

To provide a systematic shipment tracking mechanism.

To improve the consistency of delivery-status updates.

To maintain a centralized database for courier information.

1.2 Project Domain

The project belongs to the domain of web-based management systems. It combines user interface development, server-side processing, database management and basic information security.

2. BACKGROUND OF THE PROJECT

A courier operation generally consists of multiple stages: booking, verification, processing, dispatch, transportation, arrival at the destination location, out-for-delivery and final delivery. Information generated at each stage must be maintained correctly.

A computerized management system allows these stages to be represented as structured data. Instead of searching through physical registers, an authorized user can search using a tracking ID or customer information.

2.1 Real-World Application

The proposed system can be adapted for local courier offices, small logistics businesses, institutional dispatch departments and academic demonstrations. It can also serve as a foundation for a larger logistics application.

3. PROBLEM STATEMENT

The main problem is the difficulty of maintaining and retrieving courier information when operations are handled manually. Common challenges include repeated data entry, misplaced records, delayed status updates, difficulty identifying a shipment and limited reporting capability.

The project therefore proposes a centralized system that records courier information from booking to delivery and makes the current shipment status available to authorized users.

4. EXISTING SYSTEM

In a typical manual or semi-manual environment, a courier booking may be recorded in a register or spreadsheet. Tracking may depend on phone calls or manually checking records. Delivery confirmation may be maintained separately.

4.1 Drawbacks

High dependence on manual data entry.

Time-consuming search for old records.

Possibility of duplicate or inconsistent data.

Difficulty in monitoring all active shipments.

Limited automation and reporting.

Paper records may be difficult to maintain over time.

5. PROPOSED SYSTEM

The proposed Courier Management System maintains courier information in a centralized database. Each booking is associated with a unique tracking ID. The system allows authorized users to create bookings, update shipment status and confirm delivery.

5.1 Key Features

Secure login for authorized users.

Customer and courier record management.

Unique tracking ID.

Shipment-status updates.

Search and tracking facility.

Delivery confirmation.

Basic report generation.

Database-backed record storage.

6. OBJECTIVES

To develop a user-friendly courier management application.

To digitize customer and courier records.

To provide a unique tracking mechanism.

To simplify booking and delivery management.

To reduce manual paperwork.

To improve data accuracy and accessibility.

To demonstrate practical use of software engineering and database concepts.

To provide a base for future logistics-related features.

7. SCOPE OF THE PROJECT

The scope includes customer registration, courier booking, tracking ID management, shipment status updates, delivery confirmation and administrative record management.

7.1 In-Scope Features

Login and authentication

Customer records

Courier booking

Shipment tracking

Status updates

Delivery records

Basic reports

7.2 Out-of-Scope / Future Features

Live GPS tracking

Advanced route optimization

Production payment gateway

Large-scale multi-branch synchronization

AI-based delivery prediction

8. FEASIBILITY STUDY

8.1 Technical Feasibility

The project can be developed using commonly available web technologies and a relational database. The required development tools can run on a standard computer used for academic development.

8.2 Economic Feasibility

The academic prototype can be developed using freely available or educational software tools, reducing development cost.

8.3 Operational Feasibility

The system is designed with simple screens and clear operations, making it suitable for users who have basic computer knowledge.

8.4 Schedule Feasibility

The project can be developed incrementally by completing database design, interface development, backend logic, testing and documentation in separate stages.

9. REQUIREMENTS ANALYSIS

9.1 Hardware Requirements

Component	Minimum Requirement
Processor	Dual-core processor or better
RAM	4 GB or above
Storage	At least 10 GB free space
Display	Standard HD display

Network	Internet connection when required

9.2 Software Requirements

Software	Purpose
Operating System	Windows/Linux/macOS
Code Editor	Visual Studio Code or equivalent
Database	MySQL
Browser	Chrome/Edge/Firefox
Version Control	Git/GitHub – optional

10. SYSTEM USERS AND ROLES

User	Responsibilities
Administrator	Manage users, courier records, statuses and reports
Customer	Register, book courier and track shipment
Delivery Staff	Update delivery progress and confirm delivery
System/Database	Store and retrieve application data

11. SYSTEM MODULES

11.1 Login and Authentication

Validates user credentials and provides access according to the assigned role.

11.2 Customer Management

Stores customer name, phone number, address and other required information.

11.3 Courier Booking

Records sender, receiver, parcel type, weight, destination, charge and booking date.

11.4 Tracking Management

Generates or stores a unique tracking ID and provides shipment-status information.

11.5 Shipment Status

Maintains stages such as Booked, Processing, Dispatched, In Transit, Out for Delivery and Delivered.

11.6 Delivery Management

Records delivery date, delivery status and confirmation details.

11.7 Report Management

Provides summarized information for administrative monitoring.

12. SYSTEM ARCHITECTURE

The application can follow a three-layer architecture: presentation layer, application/business layer and data layer.

Layer	Responsibility
Presentation Layer	Web pages, forms, buttons, tables and tracking interface
Application Layer	Validation, business rules, authentication and status processing

Data Layer	MySQL database, tables, relationships and queries

User → Web Interface → Application Logic → Database → Application Logic → Web Interface → User

12.1 Architecture Benefits

Separation of responsibilities

Easier maintenance

Improved organization

Simpler future enhancement

Better control of database access

13. DATA FLOW DIAGRAM

The following logical flow can be used to represent the system:

Customer → Login/Booking → Courier Management → Tracking ID → Shipment Processing → Status Update → Delivery Confirmation

13.1 Level-0 DFD Description

External Entity	Process	Data Store
Customer	Book / Track Courier	Courier Database
Admin	Manage Users / Shipments	User & Shipment Database
Delivery Staff	Update Delivery Status	Delivery Status Database

13.2 Level-1 Process Flow

User submits login details.

System validates the user.

Customer enters courier details.

System validates the booking information.

System stores the courier record.

Tracking ID is assigned.

Shipment status is updated during processing.

Customer searches using tracking ID.

System displays current status.

Delivery is confirmed and final record is stored.

14. ENTITY RELATIONSHIP DESIGN

The database can be modeled using entities such as User, Customer, Courier, DeliveryStatus and Payment. A customer may create multiple courier bookings. Each courier has one unique tracking ID and may have multiple status-history entries.

14.1 Entity Relationships

User manages Courier records.

Customer creates Courier bookings.

Courier has one unique Tracking ID.

Courier can have multiple Delivery Status records over time.

Courier may have an associated Payment record.

Simplified ER structure: Customer (1) —— (M) Courier (1) —— (M) Delivery_Status

15. DATABASE DESIGN

Table	Primary Key	Important Fields
Users	user_id	name, email, password, role
Customers	customer_id	name, phone, email, address

Couriers	courier_id	tracking_id, sender, receiver, weight, destination, charge
Delivery_Status	status_id	tracking_id, status, status_date, remarks
Payments	payment_id	tracking_id, amount, payment_status

15.1 Data Validation

Tracking ID should be unique.

Required customer fields should not be empty.

Phone numbers should follow the accepted format.

Courier charge should be a valid numeric value.

Shipment status should be selected from predefined values.

16. FUNCTIONAL REQUIREMENTS

The system shall allow authorized users to log in.

The system shall allow customer information to be stored.

The system shall create courier bookings.

The system shall assign a unique tracking ID.

The system shall display shipment status.

The system shall allow authorized staff to update status.

The system shall store delivery confirmation.

The system shall allow administrators to search records.

The system shall provide basic courier reports.

17. NON-FUNCTIONAL REQUIREMENTS

Requirement	Description
Usability	Simple navigation and understandable forms
Performance	Efficient retrieval of courier records
Security	Authentication and controlled access
Reliability	Consistent storage of important records
Maintainability	Code and database should be organized
Scalability	System should support future growth
Availability	Application should be accessible when the service is running

18. USER INTERFACE DESIGN

18.1 Login Screen

The login screen should contain username/email, password and login controls. Invalid credentials should display a clear error message.

18.2 Dashboard

The dashboard can display total bookings, active shipments, delivered shipments and other useful summary information.

18.3 Booking Screen

The booking screen should collect sender, receiver, parcel and destination information. Required fields should be validated before saving.

18.4 Tracking Screen

The tracking page should provide a simple field for entering the tracking ID and display the current shipment status.

18.5 Delivery Update Screen

Authorized delivery staff can select a tracking ID and update its status with date and remarks.

19. SYSTEM WORKFLOW

User opens the application.

User logs in or registers where applicable.

Customer enters courier booking details.

System validates the submitted data.

Courier record is saved in the database.

Tracking ID is generated/assigned.

Courier is processed and dispatched.

Shipment status is updated at each stage.

Customer enters tracking ID to view progress.

Delivery staff confirms final delivery.

System stores the completed delivery record.

20. ALGORITHMS AND LOGIC

20.1 Courier Booking Logic

Receive sender and receiver details.

Receive parcel details and destination.

Validate required fields.

Calculate or enter courier charge.

Create a unique tracking ID.

Store the booking in the database.

Display booking confirmation.

20.2 Tracking Logic

Receive tracking ID from the user.

Search the courier table.

If the tracking ID exists, retrieve shipment details.

Retrieve latest status.

Display tracking information.

If no record exists, display an appropriate message.

20.3 Status Update Logic

Authorized staff selects a shipment.

System verifies the shipment record.

Staff selects the new status.

System stores status and timestamp.

Latest status becomes visible to the customer.

21. TESTING STRATEGY

Testing is performed to verify that the application works according to its requirements. Testing should cover individual functions, interactions between modules and complete user workflows.

21.1 Types of Testing

Unit Testing – Tests individual functions or modules.

Integration Testing – Tests interaction between modules.

System Testing – Tests the complete application.

Validation Testing – Checks whether input values are acceptable.

User Acceptance Testing – Confirms whether the system meets expected user needs.

22. TEST CASES

Test ID	Test Scenario	Expected Result

TC01	Valid login	User is taken to the dashboard
TC02	Invalid login	Error message is displayed
TC03	Valid courier booking	Booking is stored successfully
TC04	Missing required field	Validation message is shown
TC05	Valid tracking ID	Shipment details are displayed
TC06	Invalid tracking ID	No-record message is displayed
TC07	Status update	New shipment status is stored
TC08	Delivery confirmation	Shipment is marked as delivered
TC09	Search courier	Matching record is displayed
TC10	Unauthorized page access	Access is denied or redirected

23. SECURITY CONSIDERATIONS

Use authenticated login for protected pages.

Use role-based access for administrative operations.

Validate user inputs before database operations.

Use secure password handling in the actual implementation.

Restrict direct access to database credentials.

Use parameterized queries to reduce SQL injection risk.

Keep backup copies of important database information.

Log important administrative activities where appropriate.

24. ADVANTAGES

Reduces paperwork.

Improves record organization.

Provides faster shipment lookup.

Supports systematic status updates.

Improves visibility of courier progress.

Reduces repeated manual searching.

Provides a foundation for reporting.

Can be extended for mobile and online services.

25. LIMITATIONS

The academic prototype may not include real-time GPS tracking.

Production-level payment integration may require external services.

Actual deployment requires proper server configuration and security.

Large-scale courier operations may require distributed infrastructure.

Notification services may need third-party APIs.

26. FUTURE ENHANCEMENTS

Real-Time GPS Tracking

Integrate map and location services so authorized users can view delivery movement.

Mobile Application

Create Android/iOS applications for customers and delivery personnel.

Online Payments

Integrate a secure payment gateway for courier charges.

Notifications

Send SMS, email or app notifications when shipment status changes.

Route Optimization

Use mapping and optimization techniques to improve delivery routes.

Analytics Dashboard

Display delivery performance, booking trends and operational statistics.

Multi-Branch Support

Allow several courier branches to share data through a centralized system.

AI-Assisted Features

Future versions could explore demand forecasting or estimated delivery-time assistance.

27. EXPECTED OUTCOMES

A working prototype for courier booking and tracking.

Structured customer and shipment database.

Unique tracking mechanism.

Clear shipment status workflow.

Reduced dependency on manual records.

Improved understanding of full-stack application development.

Practical experience in database design and testing.

28. CONCLUSION

The Courier Management System is a practical academic project that demonstrates how a real-world courier process can be transformed into a structured digital workflow. The system centralizes customer, courier, tracking and delivery information and provides a clear mechanism for monitoring shipment progress.

The project also demonstrates important concepts in software engineering, web application development and database management. With additional development, the prototype can be enhanced with real-time tracking, online payments, notifications, mobile applications and analytical features.

Overall, the project provides a strong foundation for understanding how software can be designed to solve an operational problem in an organized and scalable manner.

29. REFERENCES

MySQL Documentation – SQL and relational database concepts.

MDN Web Docs – HTML, CSS and JavaScript documentation.

W3Schools – Web development reference and tutorials.

Nan Mudhalvan programme learning resources.

Standard software engineering and database management study materials.

APPENDIX A – SAMPLE COURIER STATUS VALUES

Status	Meaning
Booked	Courier booking has been created
Processing	Courier is being processed
Dispatched	Courier has left the originating facility
In Transit	Courier is moving toward the destination
Out for Delivery	Courier is with delivery personnel
Delivered	Courier has been successfully delivered
Cancelled	Booking has been cancelled where applicable

APPENDIX B – SAMPLE PROJECT OUTPUTS

The final implementation may include screenshots of the Login Page, Admin Dashboard, Customer Registration, Courier Booking Form, Tracking Page, Shipment Status Page, Delivery Update Page and Reports Page. Screenshots should be added after the application is developed and tested.